Causal Forecast-Conditioned BS-RoFormer for Ocean Sound Separation

Anonymous

Abstract

Ocean source separators normally infer a changing background only from the mixture being separated. We study whether a causal forecast of the expected background, computed from prior soundscape measurements, improves separation. The proposed system encodes historical one-third-octave levels, predicts the current background spectrum, and injects either the forecast latent state or the predicted spectrum into the band tokens of BS-RoFormer. Forecast and separation losses are optimized jointly. We specify three controlled ablations, time-disjoint deployment splits, paired SI-SDR evaluation, and a real-time factor check. Because the available public product contains hourly spectral levels rather than phase-aligned audio, the present implementation uses random-phase background surrogates. After 1,500 updates, spectrum conditioning achieves 5.74 dB SI-SDR improvement (SI-SDRi) and real-time local inference, but a matched unconditioned model reaches 6.02 dB. Thus the system passes an engineering usability gate while the claimed conditioning advantage is not supported at convergence. This distinction avoids overstating surrogate-data evidence as field performance.

1 Introduction

Passive acoustic backgrounds vary with shipping, weather, biology, and season. A separator that only observes the current mixture must estimate both the target and that background at once. Long-context soundscape forecasting offers an additional, strictly causal signal: what the background was expected to look like immediately before the mixture arrived. Our hypothesis is that this prior helps a separator reject predictable energy while retaining an unexpected injected source.

We combine the band-split RoPE Transformer (BS-RoFormer) architecture [1] with a causal recurrent forecast encoder. The contributions are: (i) a differentiable forecast–separation model; (ii) matched ablations for no forecast, latent conditioning, and spectrum conditioning; (iii) a leakage-audited deployment split; and (iv) an evaluation protocol that reports separation, forecast error, uncertainty, and computational usability.

2 Data and waveform construction

We use NOAA–Navy SanctSound hourly one-third-octave level (TOL) products [2]. MB03 deployments 01 and 02 form the training set, 03 is validation, and 04 is held-out test. A sample at hour t is eligible only when all H preceding hours and hour t are finite and exactly contiguous. Normalization statistics are fitted on the two training deployments only.

The public TOL product is an aggregate and cannot determine waveform phase. For each current spectrum S_t, we interpolate its dB envelope on FFT bins, draw a seeded random phase and Rayleigh-like magnitude jitter, invert the FFT, and normalize RMS. This gives background surrogate n_t. It preserves the coarse measured spectral envelope but not transients, phase, or within-hour events. It must not be described as recovered raw audio.

The injected signal s_t follows the controllable synthesis design developed in MINUS: harmonic partials plus band-limited noise, attack/release edges, and low-frequency amplitude modulation. Parameters and source-to-background SNR are redrawn deterministically for every sample. The mixture is $x_t = n_t + s_t$, and clipping is prevented by common rescaling.

3 Method

Let $S_{t-H:t-1} \in \mathbb{R}^{H\times F}$ be normalized historical TOL. A GRU produces

$$z_t = \mathrm{LN}(\mathrm{GRU}(S_{t-H:t-1})), \tag{1}$$

$$\widehat{S}_t = g_\theta(z_t). \tag{2}$$

Only observations ending at $t-1$ enter the encoder. The latent-conditioned model uses $c_t = z_t$; the spectrum-conditioned model uses $c_t = \widehat{S}_t$. The baseline supplies an all-zero condition and omits the forecaster.

BS-RoFormer converts mixture STFT bins into band tokens $X \in \mathbb{R}^{B\times T\times K\times D}$. Rather than concatenate an unstructured vector to audio, we learn a band-specific condition bias:

$$\widetilde{X}_{:,\tau,k,:} = X_{:,\tau,k,:} + [\mathrm{Linear}(\mathrm{LN}(c_t))]_{k,:}. \tag{3}$$

The conditioned tokens pass through the unchanged time/frequency Transformer stack and complex mask estimator. This location gives every time frame the same forecast prior while retaining frequency-specific effects.

The end-to-end objective is

$$\mathcal{L} = \mathcal{L}_{\text{wave}} + \tfrac{1}{2}\sum_r \mathcal{L}_{\text{complex STFT},r} + \lambda\|\widehat{S}_t - S_t\|_1, \tag{4}$$

with $\lambda = 0.2$ for conditioned variants. Gradients from both objectives reach the forecast encoder; this is joint training, not frozen feature export.

4 Experimental protocol

All variants use identical samples, synthesis seeds, optimizer schedule, model width, matched separator-core initialization, and checkpoint rule. The checkpoint with maximum validation SI-SDR improvement (SI-SDRi) is evaluated once on MB03_04. SI-SDR follows the scale-invariant projection form; improvement subtracts mixture SI-SDR. We also report residual-background SI-SDR, forecast MAE in dB, inference real-time factor (RTF), parameter count, and the fraction of paired test examples where conditioning beats the baseline. Metric uncertainty is a seeded paired bootstrap over test examples. BSS evaluation background is discussed by Vincent et al. [3].

Table 1: Matched 1,500-update test on 256 surrogate examples.

Condition	SI-SDRi (dB)	Δ base (dB)	Forecast MAE (dB)	RTF
None (BS-RoFormer)	6.02	0	–	.0023
Predicted spectrum	5.74	-0.28	3.43	.0024

4.1 Converged training and baseline

The practical model has 234,848 parameters and was trained for 1,500 updates with 4,000 examples per training deployment and source-to-background SNR drawn uniformly from $[-12, 12]$ dB. Spectrum conditioning reaches 5.74 dB SI-SDRi (95% interval $[5.04, 6.47]$), residual-ocean SI-SDR of 5.97 dB, and forecast MAE of 3.43 dB. The matched direct BS-RoFormer reaches 6.02 dB SI-SDRi. The paired conditional gain is -0.28 dB ($[-0.69, 0.14]$), with the conditioned model winning 35.9% of examples. A preliminary 300-update experiment had shown a +0.39 dB conditional gain; its disappearance after matched longer training demonstrates why the short run cannot support the scientific hypothesis.

Table 2: Separation by injected-source strength (128 examples per level).

Input SNR (dB)	Cond. SI-SDR (dB)	Cond. SI-SDRi (dB)	Base SI-SDRi (dB)
-12	-5.49	6.55	6.44
-6	0.32	6.32	7.05
0	5.75	5.73	6.55
+6	9.77	3.74	4.37
+12	11.89	-0.14	-0.26

At weak and equal-strength inputs the conditioned model improves the mixture by at least 5.73 dB. At +12 dB, SI-SDRi is near zero because the mixture is already target-dominant, while absolute target SI-SDR remains 11.89 dB. This motivates separate improvement and absolute-quality gates rather than one threshold for all input strengths.

5 Usability criteria

The engineering gate requires overall SI-SDRi of at least 5 dB, at least 3 dB improvement for input SNR at or below 0 dB, at least 8 dB absolute target SI-SDR for stronger inputs, and frontend RTF below one. The conditioned model passes all four checks. Its batched RTF is 0.0024; the complete chunked frontend processes a 10-second WAV at RTF 0.043 on an RTX 3060. The scientific hypothesis is supported only if a conditioned variant has positive paired SI-SDRi gain over the unconditioned baseline and its 95% interval excludes zero on unseen, phase-aligned recordings. If forecast MAE improves while separation does not, the condition is either irrelevant or poorly injected. If oracle current spectra help but forecasts do not, forecasting is the bottleneck. These outcomes are explicitly falsifying, not post-hoc exceptions.

6 Local application

The repository includes a one-megabyte checkpoint and a dependency-light local web application. A user supplies the last 24 hourly TOL rows and a WAV mixture. The application validates and normalizes history, predicts the current spectrum, separates overlapping waveform chunks, and exposes target and residual-ocean audio for playback and download. It also plots the predicted spectrum and reports duration, latency, and RTF. Requests stay on localhost; no recording or history is sent to a remote service.

7 Limitations

Hourly TOL removes phase and within-hour temporal structure. Random-phase surrogates can make spectral priors artificially direct and underestimate real ocean complexity. The current test therefore establishes implementation correctness and controlled behavior only. The decisive experiment requires timestamp-aligned raw MB03 hydrophone audio: compute historical features without crossing the test boundary, mix held-out waveform clips with the same seeded sources, freeze all choices on MB03_03, and evaluate MB03_04 once. Multiple training seeds and SNR-stratified confidence intervals should then be reported.

8 Conclusion

FC-BSR supplies a concrete, leakage-resistant route from soundscape forecasting to source separation and a usable local interface. Spectrum conditioning meets the declared surrogate benchmark across a wide intensity range, but does not beat the matched long-trained BS-RoFormer baseline. The repository is therefore an operational prototype and a falsifiable research result, not evidence that forecasts improve separation on field recordings.

References

[1] Wei-Tsung Lu, Ju-Chiang Wang, Qiuqiang Kong, and Yun-Ning Hung. Music source separation with band-split RoPE transformer. *arXiv preprint arXiv:2309.02612*, 2023.

[2] NOAA Office of National Marine Sanctuaries and U.S. Navy. SanctSound raw passive acoustic data. NOAA National Centers for Environmental Information dataset, 2020.

[3] Emmanuel Vincent, Rémi Gribonval, and Cédric Févotte. Performance measurement in blind audio source separation. *IEEE Transactions on Audio, Speech, and Language Processing*, 14(4):1462–1469, 2006.